

Fluid (Liquid-Level) Systems

Tanks, valves, and the resistance-capacitance view of fluid dynamics.

Ogata, *Modern Control Engineering*, Ch. 4.

In this tutorial you will:

- derive a single-tank model from conservation of mass,
- relate it to the first-order RC time constant, and
- extend to an interacting two-tank system.

Step through with **Ctrl+Enter**, or render a report with **publish**.

Consider a single tank of cross-sectional area C with inflow rate $q_i(t)$ and outflow through a resistive valve to atmosphere, with head (liquid level) $h(t)$. Two quantities characterize the system:

- **Capacitance** C : the change in stored liquid volume per unit change in head, $C = dV/dh$ (here just the tank's cross-sectional area for a tank with vertical sides).
- **Resistance** R : relates the outflow rate to head via the linearized relation $R = dh/dq_o$, valid for small perturbations about an operating point.

Conservation of mass (continuity) gives

$$C \frac{dh}{dt} = q_i - q_o, \quad q_o = \frac{h}{R}$$

so

$$RC\dot{h} + h = Rq_i$$

Contents

- Transfer Function of the Liquid-Level System
- Interacting Two-Tank System
- Try it yourself

Transfer Function of the Liquid-Level System

Taking the Laplace transform with zero initial conditions:

$$(RCs + 1)H(s) = RQ_i(s) \Rightarrow G(s) = \frac{H(s)}{Q_i(s)} = \frac{R}{RCs + 1}$$

This is identical in form to the electrical RC circuit (single time constant $\tau = RC$), the fluid-electrical analogy:

Fluid	Electrical
-----	-----
Flow rate, q	Current, i
Head, h	Voltage, e
Resistance, R	Resistance, R
Capacitance, C	Capacitance, C

With $R = 2 \text{ s/m}^2$ (valve resistance) and $C = 5 \text{ m}^2$ (tank area):

```
R = 2; C = 5;  
G_tank = tf(R,[R*C 1])
```

```
figure  
[hs,ts] = step(G_tank);  
plot(ts, ones(size(ts)), 'k--', ts, hs, 'b', 'LineWidth', 1.3)  
legend('Inflow step $q_i$ (before)', 'Tank level $h(t)$ (after)', 'Interpreter', 'latex', 'FontSize', 11, 'Location', 'east')
```

```

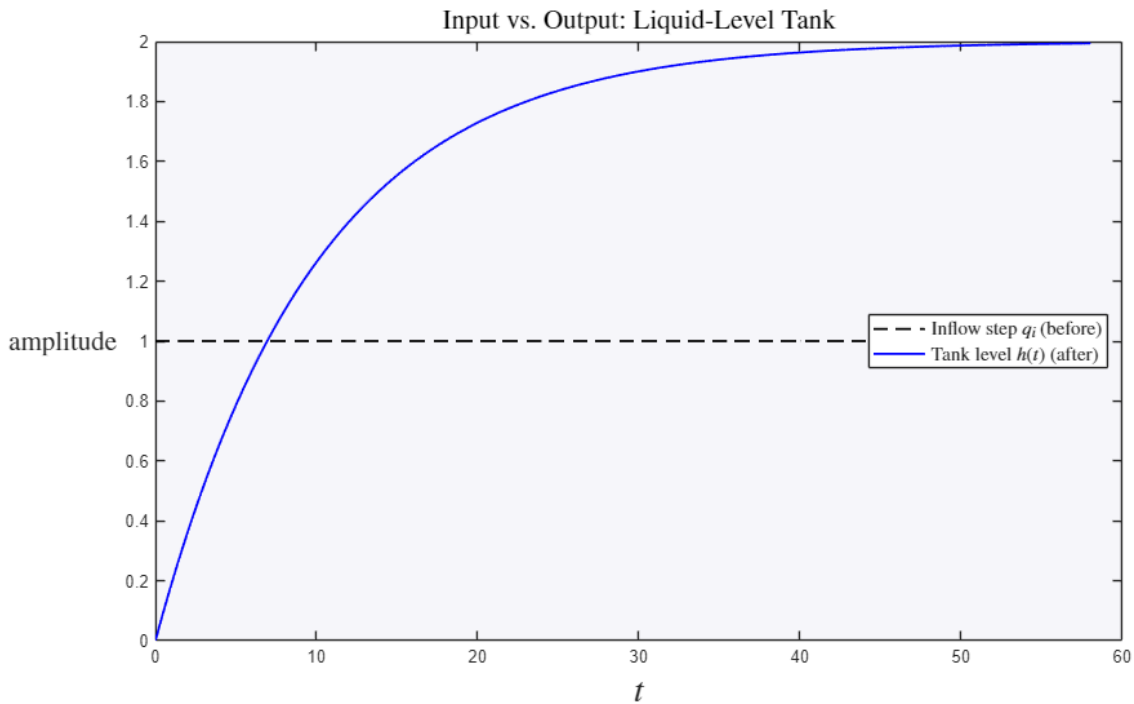
title('Input vs. Output: Liquid-Level Tank','Interpreter','latex','FontSize',16)
ylabel('amplitude','Interpreter','latex','FontSize',16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)

```

```
G_tank =
```

$$\frac{2}{10s + 1}$$

Continuous-time transfer function.



The time constant $\tau = RC = 10\text{ s}$ is the time for the level to reach 63.2% of its final value – the same first-order rule derived in `Transient and Steady-State/FirstOrder.m`.

```

tau = R*C;
disp(['Time constant tau = ' num2str(tau) ' s'])

```

Time constant tau = 10 s

Interacting Two-Tank System

For two tanks in series, each with its own resistance and capacitance, the outflow of tank 1 feeds tank 2:

$$C_1 \dot{h}_1 = q_i - \frac{h_1 - h_2}{R_1}, \quad C_2 \dot{h}_2 = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2}$$

giving the second-order transfer function

$$\frac{H_2(s)}{Q_i(s)} = \frac{R_2}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}$$

```

R1 = 1; C1 = 2; R2 = 2; C2 = 1;
num = R2;
den = [R1*C1*R2*C2, R1*C1+R2*C2+R2*C1, 1];
G_2tank = tf(num,den)

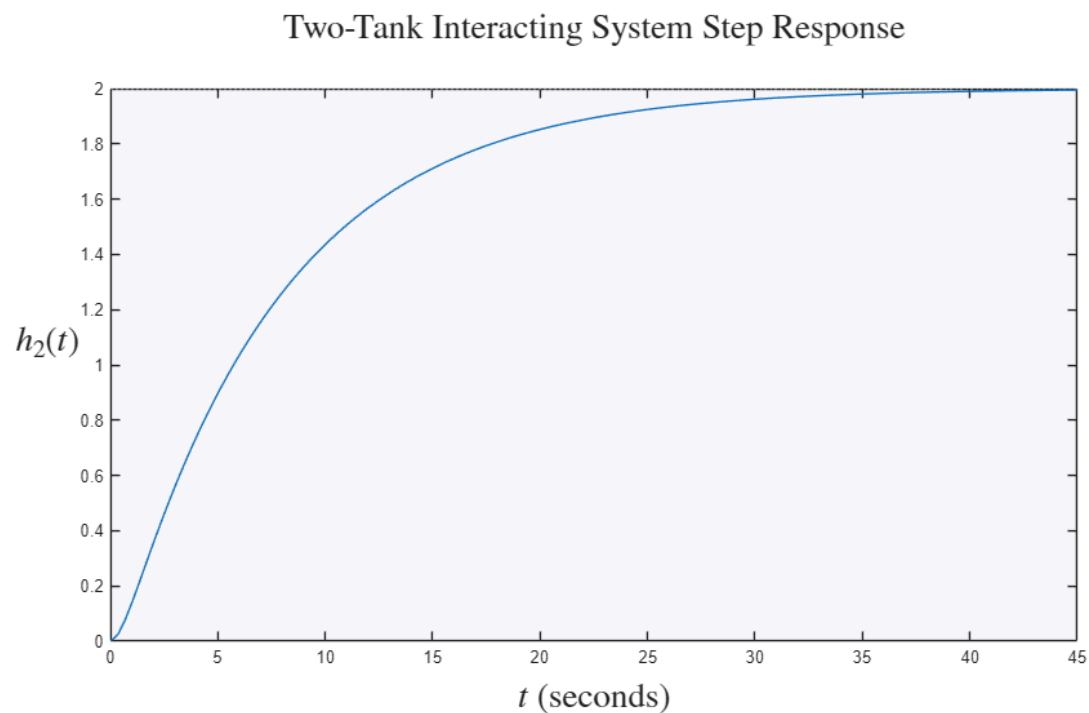
```

```
figure
step(G_2tank)
title('Two-Tank Interacting System Step Response','Interpreter','latex','FontSize',20)
ylabel('$h_2(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)
```

G_2tank =

$$\frac{2}{4s^2 + 8s + 1}$$

Continuous-time transfer function.



Try it yourself

- Double the tank area C and notice the time constant $\tau=RC$ double, so the level responds half as fast.
- Compare the two-tank **step** with a single first-order lag and see the extra S-shaped delay a second tank adds.